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Project #28: DOES ANYONE REALLY KNOW WHAT TIME IT IS?

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AIM

Problem Statement:

- Due to the increased amount of QuantiFERON®-TB Gold Plus testing, the number of timers alarming in the lab is disruptive to accessioning and leaves room for errors. Specimens are being pulled too early or left in the incubator for too long, which invalidates the testing, resulting in patients having to be called back.

Improvement (Goal) Statement:

- Since QuantiFERON®-TB Gold Plus specimens can incubate at 37°C for anywhere between 16-24 hours, if we batch them every 8 hours for removal, we can have them coming out of the incubator consistently at designated times, three times per day. Using color-coded racks, sheets, and timers would make the process visually clear, in addition to preventing incubation timing errors.

PLAN: CURRENT STATE

- QuantiFERON®-TB Gold Plus testing is a diagnostic test used to identify in vitro responses to peptide antigens associated with *Mycobacterium tuberculosis* infection. QuantiFERON®-TB Gold Plus testing results, along with patient risk assessment, radiography, and other medical and diagnostic evaluations are used to determine the possibility of *M. tuberculosis* infection in individuals. “For more than 100 years, the tuberculin skin test (TST) was the only available method for diagnosing latent tuberculosis infection. Cutaneous sensitivity to tuberculin develops from 2 to 10 weeks after infection. However, some infected individuals, including those with a wide range of conditions hindering immune functions, but also others without these conditions, do not respond to tuberculin. Conversely, some individuals who are unlikely to have *M. tuberculosis* infection exhibit sensitivity to tuberculin and have positive TST results after vaccination with Bacille Calmette-Guérin (BCG) or infection with mycobacteria other than *M. tuberculosis* complex, or undetermined other factors (1).”
- QuantiFERON®-TB Gold Plus testing has a higher degree of specificity than TST testing in the determination of *M. tuberculosis* infections by identifying cell-mediated immune responses and is now the laboratory standard for diagnosis
- Whole blood is collected in specialized QFT-Plus Blood Collection tubes, shaken to mix antigen with the blood, and then incubated at 37°C for a 16 to 24 hour incubation period
- Every time QuantiFERON®-TB Gold specimens are received, it is logged on the incubation sheet, placed in the incubator, and a timer is set for those individual specimens. 16 hours later, the timer alarms and the specimens are removed, centrifuged, and placed in the refrigerator until sent to the reference laboratory for testing. Due to the large amount of testing performed by employee health to replace TST, during the early morning, there are multiple timers that alarm constantly at multiple times.

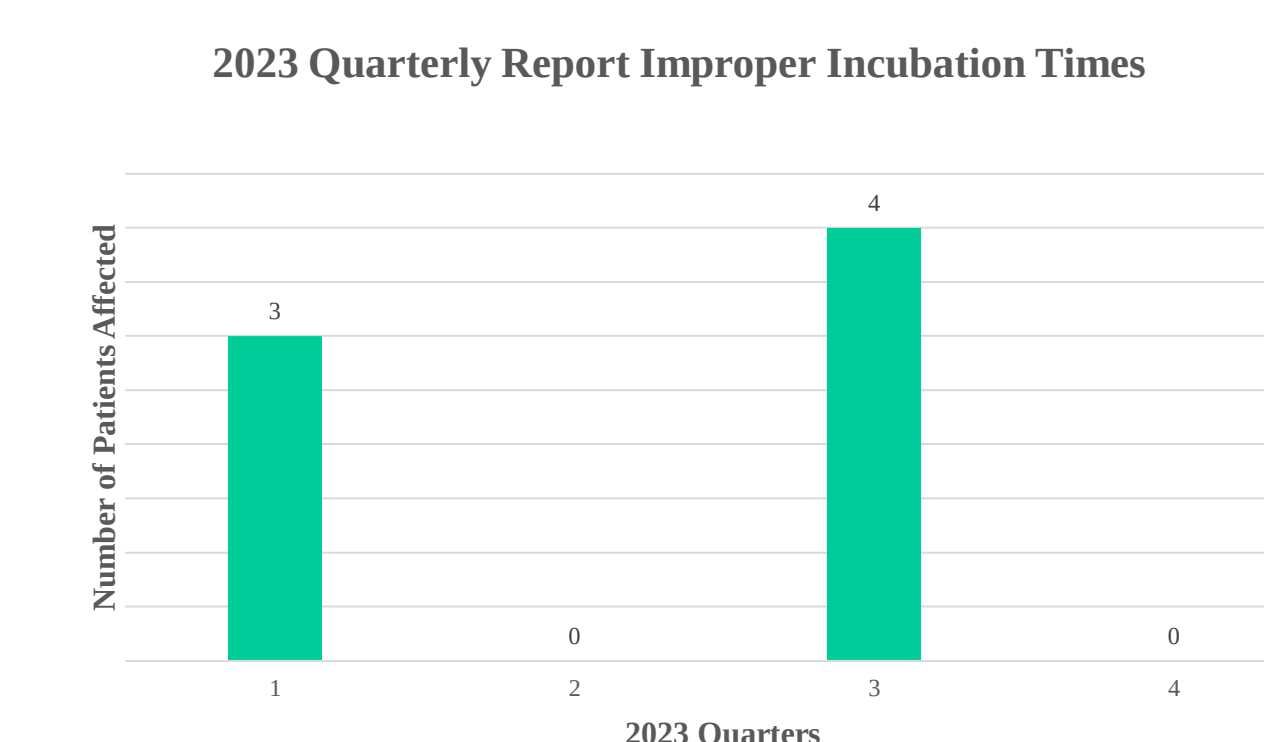
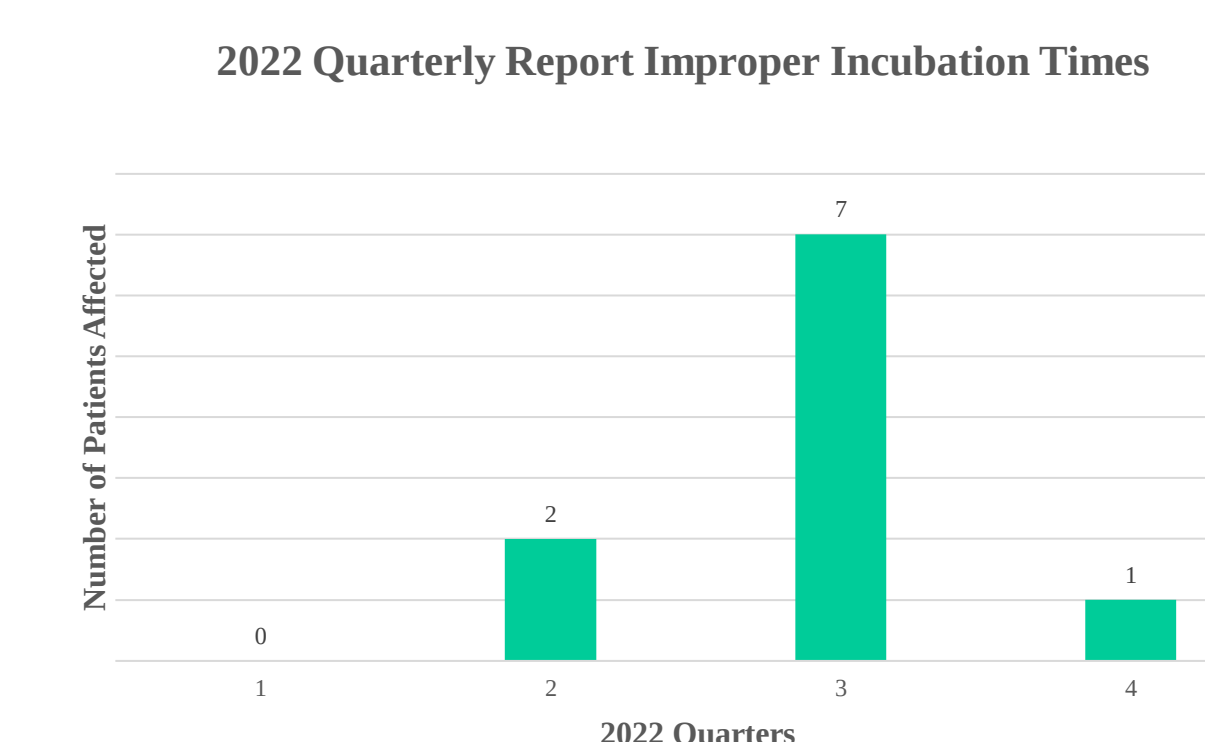
DO: CORRECTIVE ACTIONS / INTERVENTIONS

- Our volume of QuantiFERON®-TB Gold specimens has substantially increased in the last year due to a QuantiFERON®-TB Gold test drawn on every new hire and increased community exposure resulting in an increase of TB in the population and an increase in the potential of TB exposure in healthcare workers in the hospital setting
- “Before the COVID-19 pandemic, TB case counts and incidence rates had been steadily decreasing in the United States since 1992. During 2020, TB case counts and incidence rates declined substantially, likely because of factors associated with the COVID-19 pandemic. In 2021, TB case counts and incidence partially rebounded, but remained lower compared with 2019. In 2022, reported TB cases and incidence rates increased for the second year in a row (2).”
- Multiple timers alarming at multiple times lead to alarm fatigue and interruptions while multitasking, resulting in delays or omissions in removing time-dependent specimens from the incubator, resulting in the mishandling of QuantiFERON®-TB Gold specimens
- QuantiFERON®-TB Gold specimens from 0601a-1400p are placed in the yellow racks and logged on the yellow incubation sheets. Each day the yellow timer will go off at 0600a and all yellow racks will be removed, tubes spun, and placed in the fridge. QuantiFERON®-TB Gold specimens from 1401p-2200p are placed in the white rack and logged on the white incubation sheets. Each day the white timer will go off at 1400p and all white racks will be removed, tubes spun, and placed in the fridge. QuantiFERON®-TB Gold specimens from 2201p-0600a are placed in the green rack and logged on the green incubation sheets. Each day the green timer will go off at 2200p and all green racks will be removed, tubes spun, and placed in the fridge.
- Changing the way that the Laboratory processes the incubation of QuantiFERON®-TB Gold specimens will decrease the number of errors and result in fewer patient/employee callbacks for repeat draws. Getting the process done correctly the first time will result in fewer redraws and a decrease in delays for test results.
- An increase in repeat draws causes the patient/employee the inconvenience of having to come back to be redrawn, in addition to the unnecessary potential pain of a phlebotomy procedure
- The delay in getting results adds anxiety, and uncertainty to the employee and can lead to a delay in treatment, which can contribute to the additional spread of the disease

CHECK: (EVALUATION OF CHANGES)

- Initial data obtained from rL case events over two years showed that there were multiple examples of failures in the removal of QuantiFERON®-TB Gold specimens promptly resulting in patients having to be redrawn
- Since implementation in August 2023, we have not had any missed removal errors

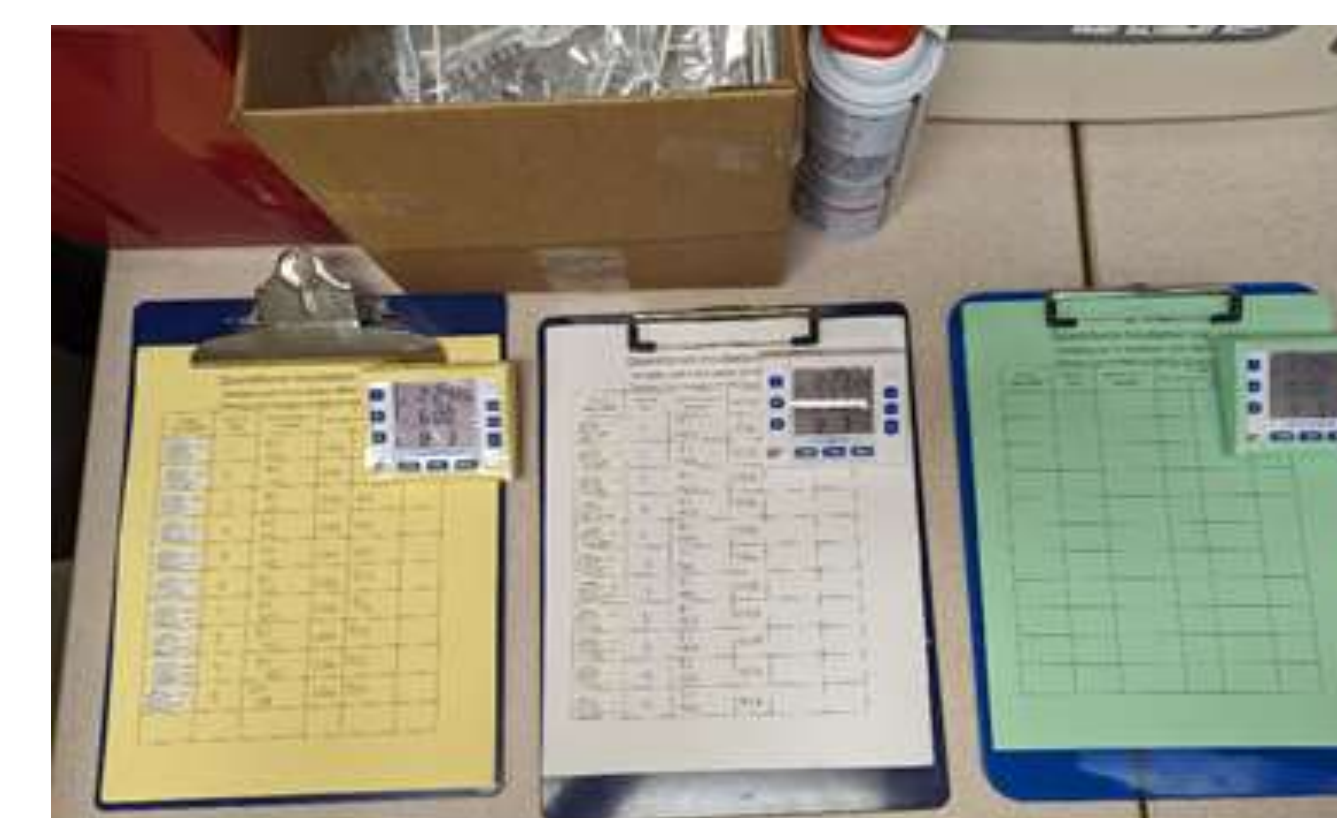
METHODS



ACT: SUSTAIN AND SPREAD

- We have seen improvements toward our goal of 0 incubation errors due to the missed removal of QuantiFERON®-TB Gold tubes from the 37°C incubator. 0 errors since implementation in August 2023
- Ongoing data collection continues, as we are only 3 months post-implementation. Laboratory team members continue to educate new hire laboratory employees and will enter deviations into rL. rL data is tabulated and reviewed quarterly. Data is reported in the monthly Laboratory Quality meeting.
- Continue to monitor the tubes sent to the Laboratory for adhering to the correct incubation process and educate those who do not follow the process, which will increase employee/patient satisfaction

TOOLS TO REDUCE IMPROPER INCUBATION TIMES



From left to right: Figure 1. Color-coded log sheets and dedicated timers that correspond to batch removal times. Figure 2. Color-coded specimen incubation racks that correspond to batch removal times.

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